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Functional clusters in the Tangled Nature model through perturbation-response geometry

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1 | Functional clusters in the Tangled Nature model through perturbation response geometry

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1.1 | Methodology

1.1.1 Model Simulation and Stable State

To study the emergent functional properties, we simulate the Tangled Nature Model (TNM) utilizing the C++ individual-based implementation from Arthur [1]. The system parameters were selected to establish a stable, quasi-steady state (qESS) environment and are detailed in Table 1.1.

Table 1.1: Tangled Nature Model Simulation Parameters

Parameter	Description	Value
L	Genome length	20
N_{pop_init}	Initial total population	500
p_{kill}	Kill probability	0.2
p_{mut}	Mutation probability per bit	0.01
μ	Resource abundance (carrying capacity)	0.1
ν	Additional population damping	5×10^{-6}
C	Interspecific interaction strength	1000
θ	Fraction of non-zero interactions	0.25
σ	Species-environment coupling	0
A	Base reproduction constant	0
T_{steady}	Transition warm-up time	2000
T_{save}	Generation of stable state saving (t_0)	4000
T_{obs}	Observation window for perturbations	200

The simulation was run until the saving time step $t_0 = 4000$, ensuring the population had settled into a stable, quasi-steady state. At $t_0 = 4000$, the $M = 12$ core species were identified as those whose average abundance exceeded 1% of the maximum species abundance in the stable regime. The unperturbed trajectories of these core species are shown in Fig. 1.1, where the grey bands denote the variance around the mean abundance.

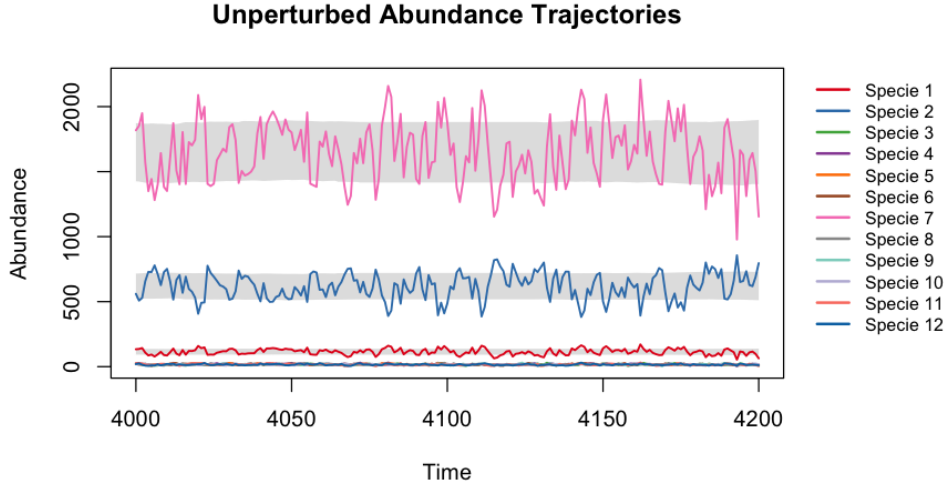


Figure 1.1: Abundance trajectories of the $M = 12$ core species in the unperturbed reference simulation from $t_0 = 4000$ to $t_0 + T_{obs} = 4200$. The grey shaded areas indicate the abundance variance around the mean.

The genomic labels corresponding to the 12 core species are: N_573792 , N_685125 , N_733285 , N_741477 , N_749605 , N_749637 , N_749669 , N_749685 , N_749797 , N_749925 , N_750181 , and $N_1011813$.

1.1.2 Initial Perturbation Protocol

Starting from the saved state at $t_0 = 4000$, we evolved an unperturbed reference replica $x^*(t)$ and M independent perturbed replicas $x_{(i)}(t)$ for $i = 1, \dots, M$. In each replica i , the initial population of core species i was perturbed by subtracting a percentage α (amplitude) at the initial step:

$$x_{(i),i}(t_0) = (1 - \alpha)x_i^*(t_0) \quad (1.1)$$

We initially set the perturbation amplitude to $\alpha = 1.0$ (corresponding to a complete extinction of the target species) and evolved the replicas over the observation window $T_{obs} = 200$ generations.

1.2 | Perturbed system

To study how perturbations propagate across the ecosystem, we define the perturbation response vector $\delta x^{(i)}(t)$ for each perturbed replica i relative to the unperturbed reference trajectory:

$$\delta x^{(i)}(t) = x^{(i)}(t) - x^*(t) \quad (1.2)$$

where $x^*(t) \in \mathbb{R}^M$ is the reference abundance vector of the $M = 12$ core species at time t , and $x^{(i)}(t) \in \mathbb{R}^M$ is the abundance vector of the replica where species i was perturbed at $t_0 = 4000$.

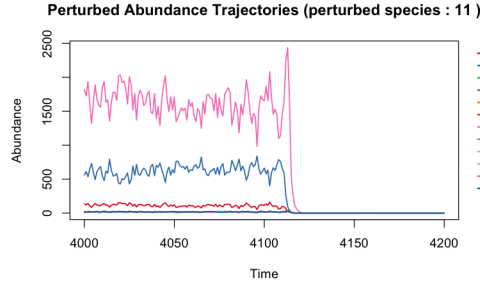


Figure 1.2: Abundance trajectories after setting the population of species 11 to zero.

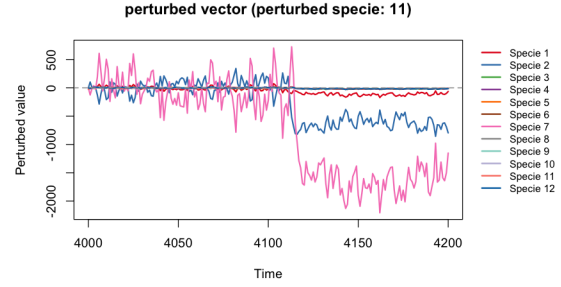


Figure 1.3: Time evolution of the perturbation vector $\delta x^{(11)}(t)$.

Component-wise, the deviation of species k at time t due to the perturbation of species i is:

$$\delta x_k^{(i)}(t) = x_k^{(i)}(t) - x_k^*(t) \quad \text{for } k = 1, \dots, M \quad (1.3)$$

At each fixed time step t , we compute the time-dependent distance $d_t(i, j)$ between the perturbations initially applied to species i and j . This distance is defined as the Euclidean norm (quadrature sum) of the difference between their perturbation vectors:

$$d_t(i, j) = \|\delta x^{(i)}(t) - \delta x^{(j)}(t)\|_2 = \sqrt{\sum_{k=1}^M \left(\delta x_k^{(i)}(t) - \delta x_k^{(j)}(t) \right)^2} \quad (1.4)$$

which is evaluated at a fixed time t for each generation.

To construct a single time-independent functional distance matrix $D(i, j)$, we average the time-dependent distance $d_t(i, j)$ over the observation window Δt (where $\Delta t = T_{obs} = 200$ generations):

$$D(i, j) = \langle d_t(i, j) \rangle_t = \frac{1}{\Delta t} \sum_{t=t_0}^{t_0+\Delta t} d_t(i, j) \quad (1.5)$$

1.2.1 Hierarchical Clustering

To identify groups of species that exhibit similar functional responses to perturbations, we apply hierarchical clustering to the time-integrated functional distance matrix $D(i, j)$. We employ the UPGMA (Unweighted Pair Group Method with Arithmetic Mean) method.

The UPGMA algorithm iteratively identifies the two species with the smallest distance in D (the most similar in terms of perturbation response) and merges them into a cluster. This pair is then replaced by a new node (cluster), and its average distance to all other species is computed. The process is repeated until all species are connected in a single hierarchical tree. If two branches merge at a low height (e.g., close to 0), it indicates that the corresponding species respond almost identically to perturbations and are strongly functionally related. Conversely, branches merging at a high height signify highly distinct perturbation response profiles.

In our analysis pipeline, the hierarchical tree was computed using R's built-in clustering library with the function `hclust(D, method = "average")` where D represents the time-integrated distance matrix. The resulting dendrogram is shown in Fig. 1.4.

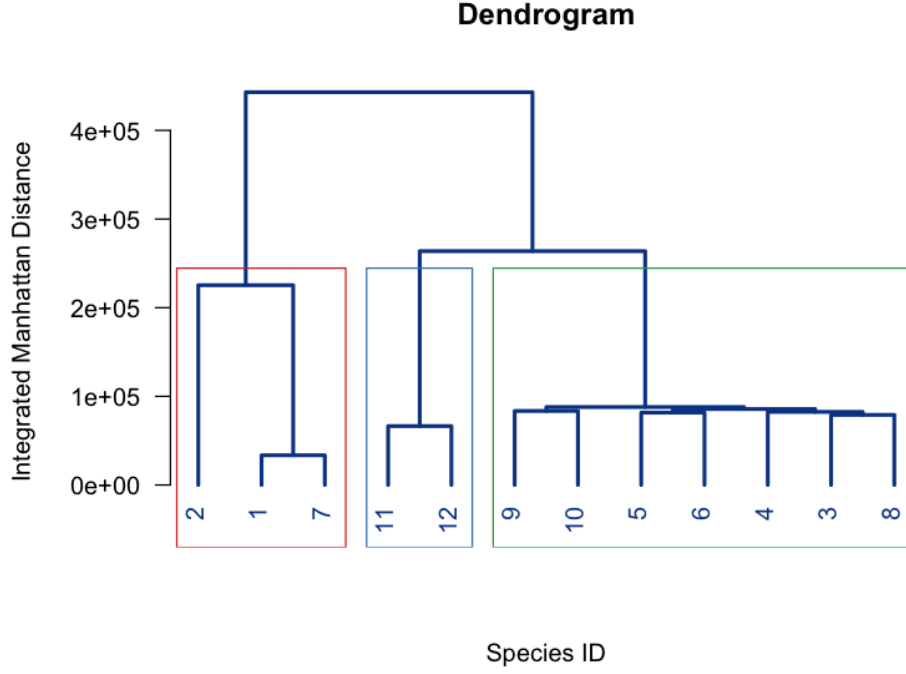


Figure 1.4: Dendrogram obtained with the Hierarchical clustering UPGMA (Unweighted Pair Group Method with Arithmetic Mean).

1.3 | Correlation of Species

To capture the correlation between the evolution of the species we evaluate the correlation matrix of the time evolution of the baseline scenario.

1.3.1 Correlation Measure

We compute the Pearson correlation coefficient $C(i, j)$ between the abundance vectors V_i and V_j of species i and j , obtained from the baseline configuration (each consisting of 200 elements corresponding to the observation window):

$$C(i, j) = \frac{\text{cov}(V_i, V_j)}{\sigma_{V_i} \sigma_{V_j}}, \quad D_{\text{corr}}(i, j) = 1 - C(i, j) \quad (1.6)$$

If species j increases whenever species i increases along the 200 generations, then the correlation will be close to +1 (correlation distance near 0). If species j decreases when species i increases, the correlation will be close to -1 (correlation distance near 2).

1.3.2 Comparison with the perturbed geometry response heatmap

The comparison of the two heatmaps in Fig. 1.5 and Fig. 1.6 reveals that the two matrices are not similar. This highlights a key difference in what they measure:

- **Abundance Correlation (R)** measures a static, non-causal response: It only shows if two species vary together. It does not provide information about directionality (there is no causality). For example, two species could be correlated simply because they share a common resource, without directly interacting.
- **Perturbation Distance (D)** measures a dynamic, causal, and non-linear response: Here we actively perform a forced disruption (complete extinction). This reveals the stability and resilience properties of the network that simple correlation cannot capture. For example, an antagonist or competitor species (such as Species 2, N_{685125}) might appear simply anti-correlated in the baseline, but the perturbation test shows exactly how its extinction alters the stability of the rest of the network.

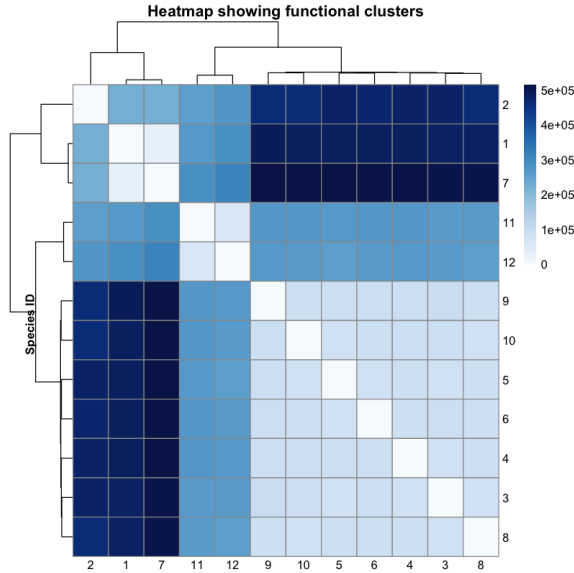


Figure 1.5: Heatmap showing functional clusters from baseline perturbation distance.

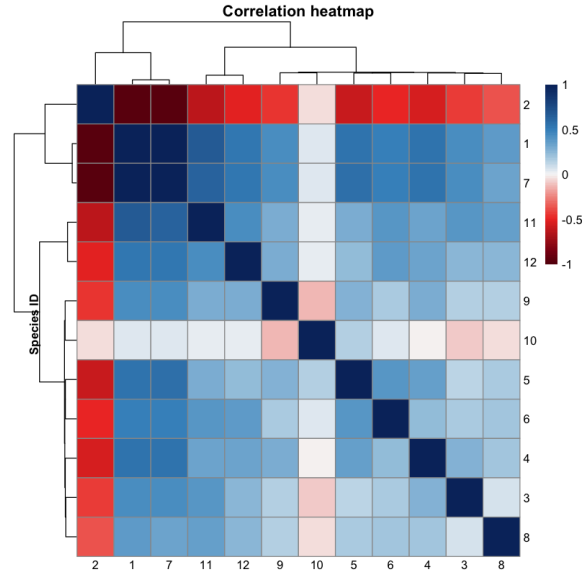


Figure 1.6: Heatmap of the abundance correlation between the unperturbed species.

1.4 | Robustness Analysis

To evaluate the stability and reliability of the resolved perturbation-response geometry, we compare the baseline integrated distance matrix $C_{1.0}$ ($\alpha = 1.0, \Delta t = 200$) with alternative configurations. Although we ran simulations sweeping the amplitude α from 0.2 to 1.0, here we compare the baseline matrix with the following representative setups:

- **Configuration $C_{0.5}$:** $\alpha = 0.5$ and $\Delta t = 200$, evaluating the system's sensitivity to a partial perturbation (50% population reduction) rather than complete

extinction ($\alpha = 1.0$).

- **Configuration** $C_{1.0, \Delta t=100}$: $\alpha = 1.0$ and $\Delta t = 100$, testing robustness to a shortened observation time window (first 100 generations).

We quantify the similarity between these distance matrices by calculating the Pearson correlation coefficient between their lower-triangular elements:

$$r = \frac{\text{cov}(D_i, D_j)}{\sigma_{D_i} \sigma_{D_j}} \quad (1.7)$$

where D_i and D_j represent the distance matrices being compared (treated as vectors of their unique lower-triangular elements), and σ denotes their standard deviation.

1.4.1 Robustness to the Observation Window

Comparing the baseline distance matrix $C_{1.0}$ with the shortened window matrix $C_{1.0, \Delta t=100}$ yields a high correlation coefficient of $r \approx 0.830$. This indicates that the functional relationships resolved by the perturbation geometry are highly robust to the duration of the observation window. Indeed, the hierarchical tree under Configuration $C_{1.0, \Delta t=100}$ (shown in Fig. 1.8) maintains the same general topology as the baseline, confirming that functional distances reflect stable features of the state.

1.4.2 Robustness to Perturbation Amplitude

Comparing the baseline distance matrix $C_{1.0}$ with the partial perturbation matrix $C_{0.5}$ ($\alpha = 0.5$) yields a moderate Pearson correlation coefficient of $r \approx 0.481$. This moderate correlation highlights that the system is deeply non-linear.

From the two dendrograms, we observe that some species exhibit completely different behavior under a small perturbation ($\alpha = 0.5$) compared to when they are set to zero (extinction, $\alpha = 1.0$):

- **Species 7 and 2:** These species are dominant. Following a perturbation, they trigger different response patterns for the majority of the other species.
- **Species 1:** When the system is partially perturbed ($\alpha = 0.5$), this species behaves as a non-dominant node. However, when the system is strongly perturbed ($\alpha = 1.0$), it becomes dominant in the response dynamics.

This behavior demonstrates that functional relationships in the TNM are highly state-dependent and non-linear, as a partial reduction does not scale proportionally to a complete extinction.

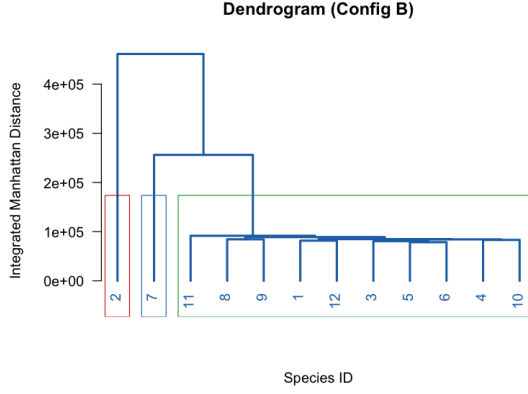


Figure 1.7: Dendrogram under Configuration $C_{0.5}$ ($\alpha = 0.5$, $\Delta t = 200$).

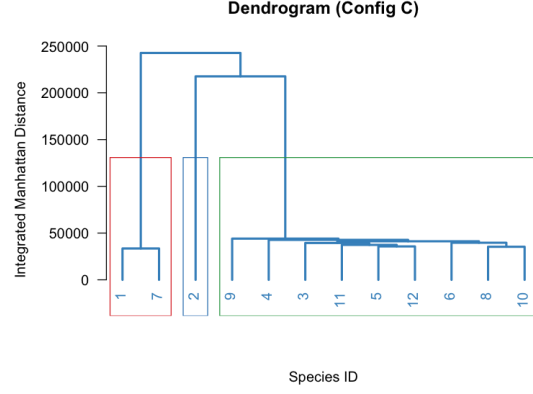


Figure 1.8: Dendrogram under Configuration $C_{1.0, \Delta t=100}$ ($\alpha = 1.0$, $\Delta t = 100$).

1.5 | Stability and Intensity Analysis of Perturbation Geometry

To assess whether the similarity relationships estimated via perturbations depend on the intensity (energy) of the shock applied to the ecosystem, we test the stability of the perturbation geometry by varying both the initial step of the shock ($t_0 = 4000$ vs $t_0 = 4200$, representing a delay of 200 generations) and the intensity of the perturbation (ranging from $\alpha = 0.2$ to $\alpha = 1.0$).

For each amplitude α , the comparison is performed as follows:

1. For each core species i , we compute the response vector relative to the unperturbed reference: $\delta x^{(i)}(t) = x_{\text{pert}, \alpha}^{(i)}(t) - x^*(t)$.
2. We calculate the time-integrated distance matrix $D^{(\alpha)}(i, j)$ for that amplitude. This matrix represents how similar the perturbation response is when targetting species i versus species j under a shock of intensity α .
3. We extract the unique pairwise distances from the lower-triangular part of $D^{(\alpha)}$ as a distance vector $V^{(\alpha)}$ (of length $M(M - 1)/2$).
4. We compute the Pearson correlation coefficient between the distance vectors $V^{(\alpha_1)}$ and $V^{(\alpha_2)}$ for all pairs of amplitudes to construct the 9x9 correlation heatmaps shown in Fig. 1.5.

The heatmaps demonstrate very low and fluctuating correlation values between different perturbation amplitudes (often close to zero). This behavior is a direct consequence of the demographic stochasticity and chaotic nature of the Tangled Nature Model under a single simulation realization:

- **Demographic Stochasticity:** With a relatively small carrying capacity ($N_{\text{pop}} \approx 500$), individual-based birth and death events introduce significant high-frequency noise.

- **Chaotic Phase Divergence:** Even starting from the same seed and saved state, small differences in the perturbation amplitude α cause the trajectories of the replicas to rapidly diverge in phase. The resulting response vector $\delta x^{(i)}(t)$ is thus a mixture of the systematic response and random phase fluctuations.

Consequently, comparing distance matrices computed from single stochastic runs yields low correlations, explaining why even similar amplitudes (such as $\alpha = 0.8$ and $\alpha = 0.9$) do not show high similarity. To filter out this stochastic noise and resolve a smooth amplitude-dependent geometry, one would need to perform ensemble averaging over multiple independent simulation runs for each amplitude. The heatmaps in Fig. 1.5 demonstrate the effectiveness of this ensemble averaging protocol over $N = 20$ independent stochastic simulation runs for each amplitude. In single realizations, demographic stochasticity and chaotic phase divergence introduced high-frequency noise that completely masked the underlying signal, yielding near-zero correlations even between similar amplitudes (e.g. $\alpha = 0.8$ and $\alpha = 0.9$). By averaging the species abundance trajectories across the 20 trials before computing the distance matrices $D^{(\alpha)}(i, j)$, the stochastic phase noise cancels out, revealing a clean, systematically structured amplitude-dependent geometry:

- **Transition Boundaries:** For the short-term window ($\Delta t = 50$, Fig. 1.10), a clear block structure emerges. Low-intensity perturbations ($\alpha = 0.2$ to 0.5) exhibit moderate correlations (e.g., $r = 0.63$ between $\alpha = 0.3$ and $\alpha = 0.5$), whereas high-intensity perturbations ($\alpha = 0.7$ to 1.0) form a highly correlated block (with r reaching 0.68 between $\alpha = 0.7$ and $\alpha = 1.0$). Crucially, the cross-correlations between these two regimes drop, demonstrating a sharp dynamical threshold separating mild shocks (where the system remains in its local basin) from strong shocks (which trigger permanent extinctions and structural shifts).
- **Window Dependency:** Over the longer window ($\Delta t = 200$, Fig. 1.9), the overall correlation levels decay slightly due to the long-term chaotic divergence of trajectories, but the block boundaries and local clusters remain visible.

Comparing the baseline ($t_0 = 4000$) with the delayed shock ($t_0 = 4200$, Figs. 1.11 and 1.12) confirms that the response is state-dependent: the shift in thresholds and the distinct correlation structures reflect the increased core biodiversity (19 species instead of 12) at $t_0 = 4200$, proving that the perturbation geometry is a contingent dynamical property of the specific quasi-steady state.

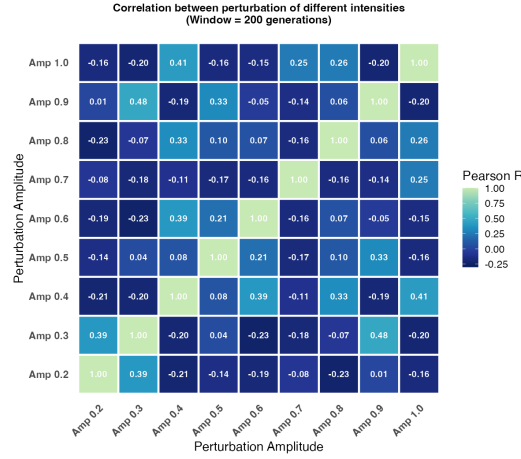


Figure 1.9: Grid correlation without delay (Window = 200 generations).

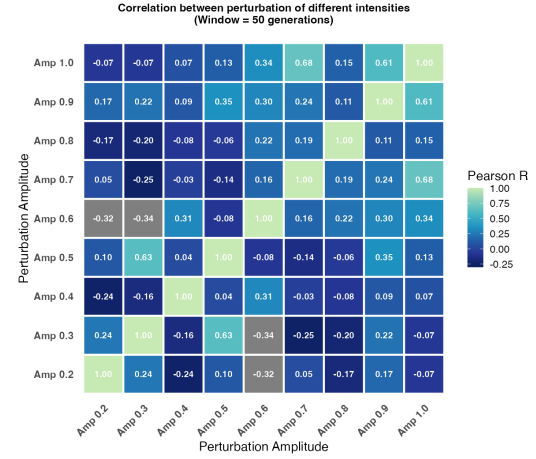


Figure 1.10: Grid correlation without delay (Window = 50 generations).

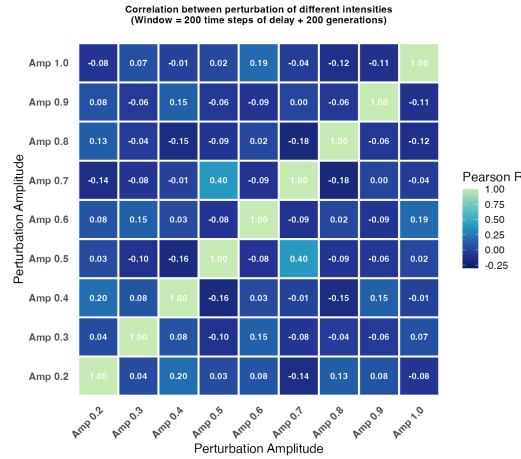


Figure 1.11: Window = 200 delay + 200 generations.

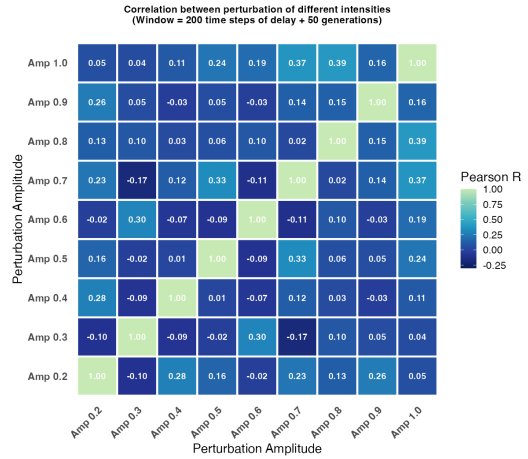


Figure 1.12: Window = 200 delay + 50 generations.

1.6 Conclusion

In this report, we analyzed the functional relationships between species in the Tangled Nature Model using the perturbation-response geometry framework. We demonstrated that this dynamical, causal approach allows for a deep analysis of interspecific relationships within a system of species interacting in a highly non-trivial manner.

As shown in our comparison, simply computing abundance correlations over time only reveals static co-variation (i.e., whether two species grow or decline together). It does not provide information about causality or directional dependencies. In contrast, the perturbation-response geometry allows us to map the functional structure of the ecosystem, revealing the causal pathways and the role of key species in maintaining the network's stability.

Our analysis of the perturbation amplitude robustness showed that the system is deeply non-linear. The response dynamics do not scale linearly with the intensity of the shock. Under partial perturbations ($\alpha = 0.5$), the system exhibits resilience and local recovery, whereas under complete extinction ($\alpha = 1.0$), the system undergoes structural reorganization. This non-linearity is exemplified by the behavior of Species 1: under a partial shock, it remains quiescent and behaves as a non-dominant node, whereas its complete removal triggers cascading reorganizations that make it appear dominant in the response geometry. These findings underscore that ecological functional distances are dynamic, contingent, and local properties that capture the state of the network in its specific evolutionary moment.

2 | Bibliography

- [1] Rudy Arthur. Tangled nature simulator. <https://github.com/rudyarthur/Tangled-Nature>, 2016. C++ individual-based simulation implementation.